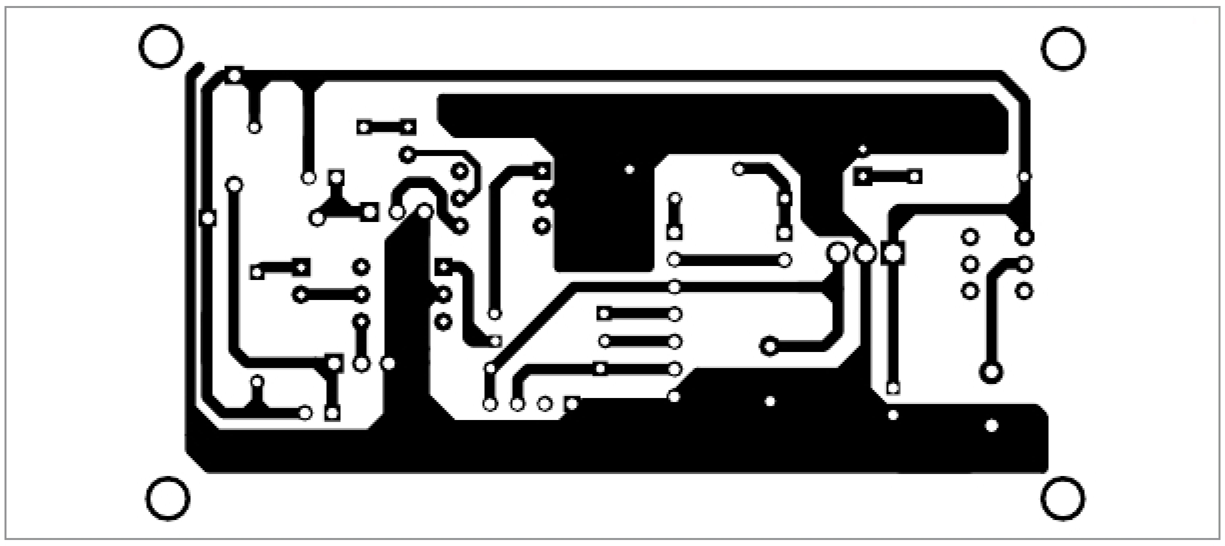

Fig. 5: Actual-size PCB layout of ESP8266-based home automation system

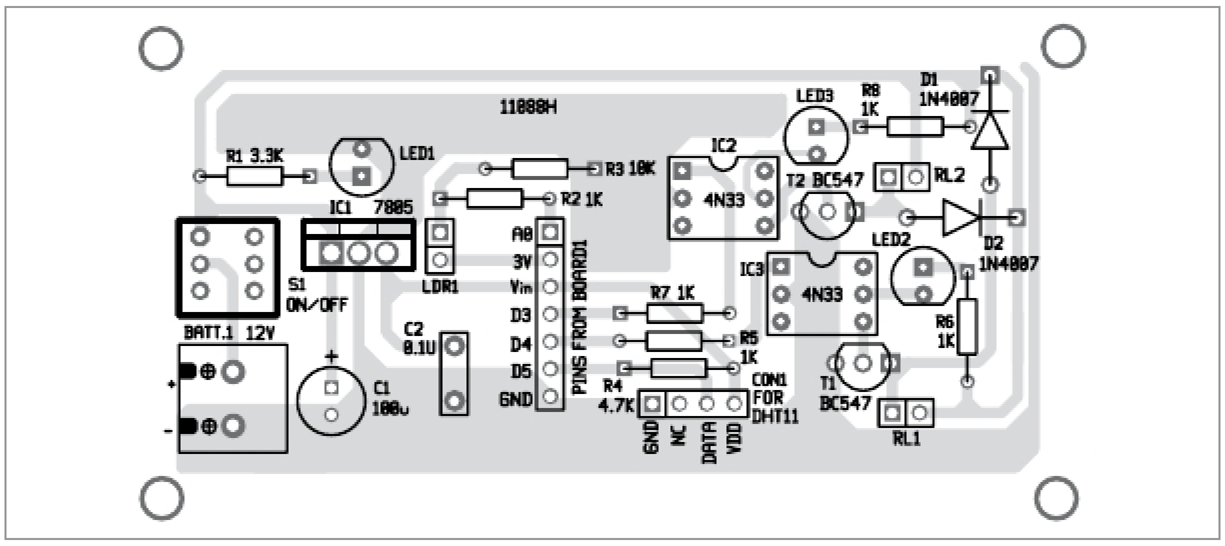

Fig. 6: Components layout for the PCB

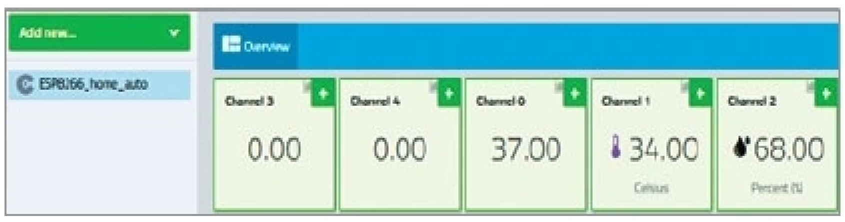

Fig. 7: Overview of channels

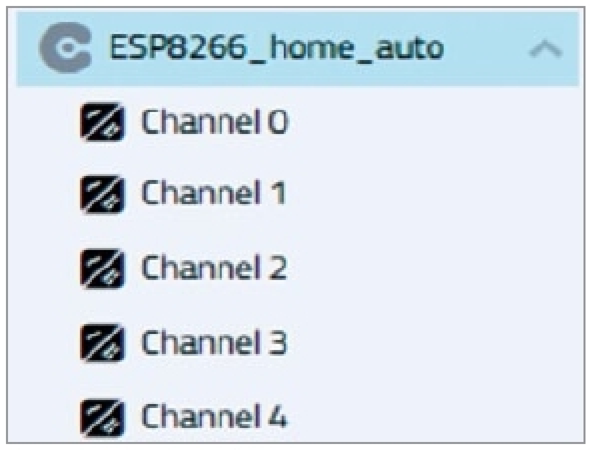

Fig. 8: Channels on Add New page

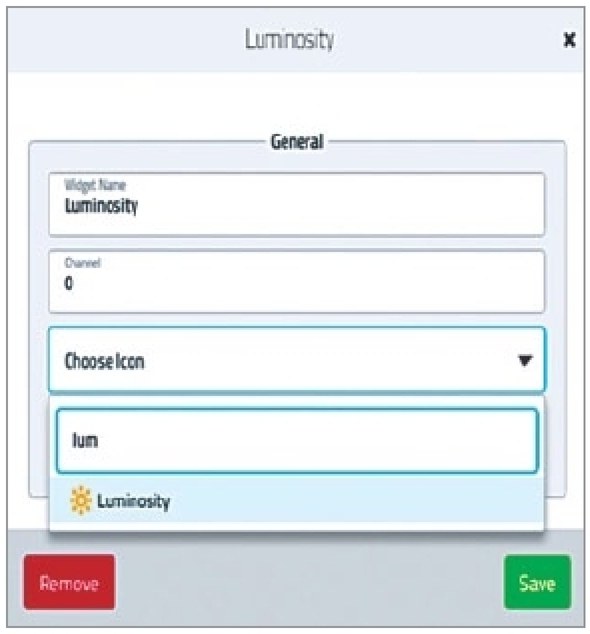

Fig. 9: Channel 0 settings

Fig. 10: Channel 0 dashboard

For programming NodeMCU module, three unique identities are required from Cayenne website, namely, MQTT user name, MQTT password and Client ID. When the device is connected to the network, these IDs help Cayenne website to find the device and start communicating with it.

Other IDs required are your Wi-Fi SSID and password, if any, to connect with the local Wi-Fi network.

Before compiling and loading the code to NodeMCU module, install the following libraries in Arduino IDE as described below.

Cayenne-MQTT-ESP8266 library. This allows NodeMCU to communicate with devices on Cayenne platform.

DHT library. This allows DHT11 sensor to communicate with NodeMCU. Search Unified under Library Manager and install Adafruit Unified Sensor.

To add the above two libraries to Arduino IDE, go to Sketch→Include Library→Add .ZIP Library… Locate and install the files.

Alternatively, you can directly download and copy these libraries to Arduino/library folder of Arduino IDE. Download the libraries from *https://github.com/myDevicesIoT/Cayenne-MQTT-ESP* and *https://github.com/adafruit/DHT-sensor-library*

Construction and testing

An actual-size PCB layout for ESP8266-based home automation system is shown in Fig. 5 and its components layout in Fig. 6. Assemble the components on the PCB as per circuit diagram. CON2-CON5 are not shown in the PCB (can be connected externally).

Building dashboard on Cayenne IoT platform

Open website *https://mydevices.com* and create an account through Sign Up Free button on the top-right. Click on Bring Your Own Thing. The next page will give three unique strings: MQTT Username, MQTT Password and Client ID.

Copy these strings and paste in appropriate fields in Arduino code (ESP8266_HA.ino). Then, load the code in NodeMCU module through a USB cable. Wait for some time on this page to connect with the device on the network.

Once the module is connected to Wi-Fi, the Web page will move to the next screen. Here, your device will be given a unique name, which can be changed through Settings on the top-